Introduction to Artificial Intelligence
23CSAMAJ301

Unit 1: Introduction to AI
1. Intelligent Agents: Agents and environment, the concept of Rationality, the nature of environment, the structure of Agents. 2. Knowledge-Based Agents**:** Introduction to Knowledge-Based Agents, The Wumpus World. 3. Problem-solving: Problem-solving agents.

What is AI?

"AI" stands for Artificial Intelligence. It refers to the ability of computer systems to perform tasks that typically require human intelligence, such as learning, problem-solving, and decision-making. AI encompasses a wide range of technologies and approaches that enable machines to mimic human cognitive functions.

- **Mimicking Human Intelligence:**

  AI aims to create machines that can perform tasks that humans usually need to be intelligent for. This includes things like understanding language, recognizing images, solving problems, and making decisions.

- **Learning and Adaptation:**

  A key aspect of AI is its ability to learn from data and improve its performance over time. This is often achieved through machine learning algorithms that allow AI systems to identify patterns and make predictions based on the information they are given.

- **Wide Range of Applications:**

  AI is used in many different fields, from healthcare and finance to transportation and entertainment. Examples include virtual assistants, fraud detection, self-driving cars, and personalized recommendations.

- **Not Human Intelligence:**
  It's important to remember that AI is not the same as human intelligence. While AI systems can perform intelligent tasks, they don't think or feel like humans do.

In essence, AI is a broad field focused on creating intelligent machines that can assist with or automate a wide range of tasks that previously required human intellect.

<u>Intelligent Agents: Agents and environment</u>

In the context of Artificial Intelligence (AI), an intelligent agent is a system that can perceive its environment, make decisions, and take actions to achieve specific goals. The environment is the context in which the agent operates, providing the setting for its actions and observations. Essentially, AI agents are designed to interact with their surroundings and learn from those interactions to improve their performance.

In other words:

Intelligent Agents:

- Definition:

  An AI agent is any entity that can perceive its environment through sensors and act upon it using effectors (or actuators).

- Function:

  Agents perceive the environment, make decisions based on that perception, and perform actions to achieve their goals.

- Examples:
  - Human: Has sensors (eyes, ears, etc.) and effectors (limbs, mouth, etc.).
  - Robot: Uses cameras, infrared sensors, and motors.
  - Software: Uses encoded bit strings as programs and actions.
  - Virtual assistants: (Siri, Alexa)
  - Autonomous vehicles:
  - Spam filters:

Rationality:

- A rational agent is one that makes the best possible decisions based on its knowledge and the information it has received, aiming to maximize its performance measure.
- Rationality depends on four key factors: the agent's performance measure, its prior knowledge of the environment, the actions it can perform, and its history of perceptions.
- For example, a chess-playing agent is rational if it makes moves that lead to winning the game, given its knowledge of the game and the current board state.
- Perfect rationality assumes unlimited computational resources and perfect knowledge, which is often unrealistic, but it serves as a benchmark for evaluating AI agents.

In essence, AI is about creating rational agents that can intelligently interact with their environment to achieve desired outcomes.

The nature of environment:

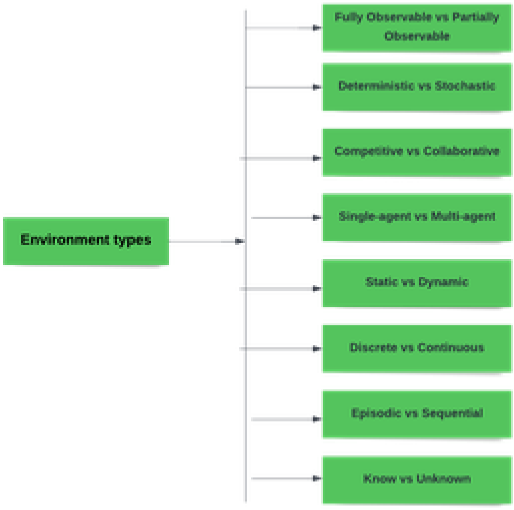
In artificial intelligence (AI), the environment is the context in which an AI agent operates and makes decisions. Understanding the nature of the environment is crucial for designing and developing effective AI systems, as it dictates how an agent perceives, acts, and learns. Environments can be classified based on several characteristics, including observability, determinism, dynamics, and the number of agents involved.

Types of Environments in AI

An environment in artificial intelligence is the surrounding of the agent. The agent takes input from the environment through sensors and delivers the output to the environment through actuators. There are several types of environments:

- Fully Observable vs Partially Observable
- Deterministic vs Stochastic
- Competitive vs Collaborative
- Single-agent vs Multi-agent
- Static vs Dynamic
- Discrete vs Continuous
- Episodic vs Sequential
- Known vs Unknown

Environment types



1. Fully Observable vs Partially Observable

- When an agent sensor is capable to sense or access the complete state of an agent at each point in time, it is said to be a fully observable environment else it is partially observable.
- Maintaining a fully observable environment is easy as there is no need to keep track of the history of the surrounding.
- An environment is called **unobservable** when the agent has no sensors in all environments.
- **Examples:**
    - **Chess -** the board is fully observable, and so are the opponent's moves.
    - **Driving -** the environment is partially observable because what's around the corner is not known.

2. Deterministic vs Stochastic

- When a uniqueness in the agent's current state completely determines the next state of the agent, the environment is said to be deterministic.
- The stochastic environment is random in nature which is not unique and cannot be completely determined by the agent.
- **Examples:**
    - **Chess -** there would be only a few possible moves for a chess piece at the current state and these moves can be determined.
    - **Self-Driving Cars-** the actions of a self-driving car are not unique, it varies time to time.

3. Competitive vs Collaborative

- An agent is said to be in a competitive environment when it competes against another agent to optimize the output.
- The game of chess is competitive as the agents compete with each other to win the game which is the output.
- An agent is said to be in a collaborative environment when multiple agents cooperate to produce the desired output.
- When multiple self-driving cars are found on the roads, they cooperate with each other to avoid collisions and reach their destination which is the output desired.

4. Single-agent vs Multi-agent

- An environment consisting of only one agent is said to be a single-agent environment.
- A person left alone in a maze is an example of the single-agent system.
- An environment involving more than one agent is a multi-agent environment.
- The game of football is multi-agent as it involves 11 players in each team.

5. Dynamic vs Static

- An environment that keeps constantly changing itself when the agent is up with some action is said to be dynamic.
- A roller coaster ride is dynamic as it is set in motion and the environment keeps changing every instant.
- An idle environment with no change in its state is called a static environment.
- An empty house is static as there's no change in the surroundings when an agent enters.

6. Discrete vs Continuous

- If an environment consists of a finite number of actions that can be deliberated in the environment to obtain the output, it is said to be a discrete environment.
- The game of chess is discrete as it has only a finite number of moves. The number of moves might vary with every game, but still, it's finite.
- The environment in which the actions are performed cannot be numbered i.e. is not discrete, is said to be continuous.
- Self-driving cars are an example of continuous environments as their actions are driving, parking, etc. which cannot be numbered.

7.Episodic vs Sequential

- In **an Episodic task environment**, each of the agent's actions is divided into atomic incidents or episodes. There is no dependency between current and previous incidents. In each incident, an agent receives input from the environment and then performs the corresponding action.
- **Example:** Consider an example of **Pick and Place robot**, which is used to detect defective parts from the conveyor belts. Here, every time robot(agent) will make the decision on the current part i.e. there is no dependency between current and previous decisions.

- In a **Sequential environment**, the previous decisions can affect all future decisions. The next action of the agent depends on what action he has taken previously and what action he is supposed to take in the future.
- **Example:**
    - **Checkers-** Where the previous move can affect all the following moves.

8. Known vs Unknown

- In a known environment, the output for all probable actions is given. Obviously, in case of unknown environment, for an agent to make a decision, it has to gain knowledge about how the environment works.

Structure of Agent in AI: Components, Types, and How They Work

The structure of an agent in AI explains how intelligent systems observe their surroundings, make decisions, and take action. At its core, it includes sensors to gather data, actuators to interact with the environment, and a function that decides what to do based on what it senses. Different types of agents, like reflex-based, goal-driven, or learning agents, handle tasks in unique ways. Their design depends heavily on the environment they operate in. Understanding this structure helps developers build smarter, more efficient AI systems suited for real-world use, from self-driving cars to recommendation engines and smart home devices.

What is an Agent in AI?

An agent in artificial intelligence is any entity that can perceive its environment through sensors and act upon it using actuators. Think of it as a loop: it sees or senses something, decides what to do, and then acts.

Here's a simple analogy: a self-driving car is a classic example of an AI agent. Its cameras and radar act as sensors. Its steering, acceleration, and braking systems act as actuators. It's constantly analyzing what's around it, other cars, pedestrians, and road signs, and choosing the next best move. That loop of sensing, deciding, and acting is what makes it an agent.

The key idea is: an AI agent is autonomous. It can make decisions on its own, based on the data it perceives.

What is the Structure of an Agent in AI?

The structure of an agent in AI is the foundation of how it works. It's not just about what the agent does, but how it processes the world around it to take action. At its core, every agent has a **few essential components:**

1. Sensors

These are the agent's eyes and ears, what it uses to perceive the environment. Depending on the system, sensors could be:

Cameras in a robot

Microphones in a voice assistant

Temperature detectors in a smart thermostat

Web data inputs into a recommendation engine

Sensors gather raw data about the state of the world.

2. Actuators

These are the parts that let the agent act. In physical systems, they could be motors, servos, or mechanical parts. In software systems, actions might include sending alerts, modifying data, or triggering other digital processes.

If a thermostat senses that a room is too cold, the actuator is what actually turns on the heater.

3. Percept Sequence

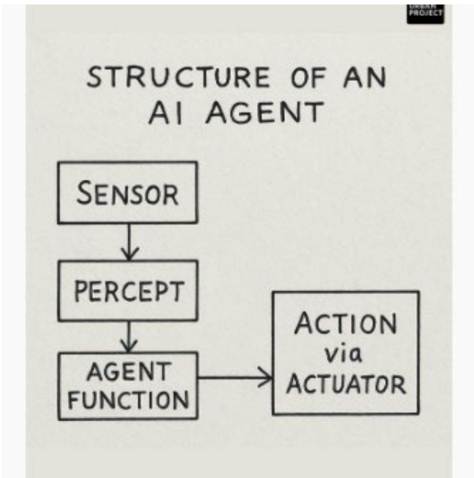
This is essentially the history of everything the agent has perceived so far. Rather than working on just the latest data point, some agents need to analyze trends over time, recognize patterns, or understand cause-effect relationships that emerge across multiple events.

Not every agent uses its full percept sequence, but more complex ones definitely do.

4. Agent Function

This is the real decision-maker. The agent function takes the percept (or percept sequence) and maps it to a specific action. It's the logic or strategy that says, "Given what I'm sensing, what should I do?"

You could think of the agent function as the brain of the operation, though it can range from something super simple (like a rule-based system) to something highly complex and adaptive (like a deep learning model).



What Are the Main Types of AI Agents?

This is where it gets interesting. Not all AI agents are built the same. Depending on the complexity of the environment and the task at hand, different types of agents are used.

1. What is a Simple Reflex Agent?

These agents operate on IF-THEN rules. They don’t remember past actions or look ahead. They just act based on the current situation.

Behavior: “If I see dirt, start vacuuming.”
No memory or planning
Only responds to current percept
Example: A basic vacuum cleaner robot. It detects dirt → it moves. That’s it. If it hits a wall, it turns.

Simple? Yes. Effective in limited scenarios? Absolutely.

2. What is a Model-Based Reflex Agent?

These agents are a step up. They maintain some kind of internal state that represents part of the world they can’t directly sense at the moment.
This means they can handle situations where not everything is visible all the time.

- Keeps a model of the world
- Updates internal state as new data comes in

Example: A home security system. If a window sensor is triggered and no one disarmed the alarm, it assumes a break-in. It remembers past states (armed, disarmed, motion detected) and acts accordingly.

3. What is a Goal-Based Agent?

Here's where decision-making gets more sophisticated. A goal-based agent doesn't just react; it evaluates whether an action will help it achieve a goal.

- Plans ahead
- Considers different outcomes
- Chooses based on what gets it closer to its goal

Example: A chess-playing AI. It doesn't just move a piece because it can, it simulates potential future moves and picks the one most likely to lead to a win.

These agents require more computing power but offer far more intelligent behavior.

4. What is a Utility-Based Agent?

While goal-based agents focus on reaching a specific outcome, utility-based agents go a level deeper. They're not just trying to "win the game", they want to win in the best possible way.

A utility-based agent doesn't just ask "Does this action achieve my goal?" It asks, "How good is this outcome compared to all other possible outcomes?"

- Evaluates multiple outcomes
- Uses a utility function to rank them
- Chooses the action with the highest utility

Example: A self-driving car.

Let's say the goal is to reach a destination. That could be done by taking the shortest route, but what if that road is congested or has a high accident rate? A utility-based agent factors in safety, speed, fuel efficiency, and maybe even passenger comfort, and picks the best route accordingly.

It's about maximizing value, not just hitting a target.

5. What is a Learning Agent?

This is where the magic really happens.

A learning agent is capable of improving its behavior over time by learning from experience. Instead of being pre-programmed with fixed rules, it adjusts its strategies based on feedback.

These agents typically have four components:

- Learning Element – This is what makes improvements based on performance.
- Performance Element – Responsible for selecting actions.
- Critic – Gives feedback about how the agent is doing relative to a standard.
- Problem Generator – Suggests exploratory actions to improve learning.

Example: Think of how recommendation engines like Netflix or YouTube work. They learn from what you watch, how long you watch, when you skip, and start adjusting what they show you next. That's learning in action.

These systems are especially useful in environments where things change over time or where the full set of rules isn't known upfront.

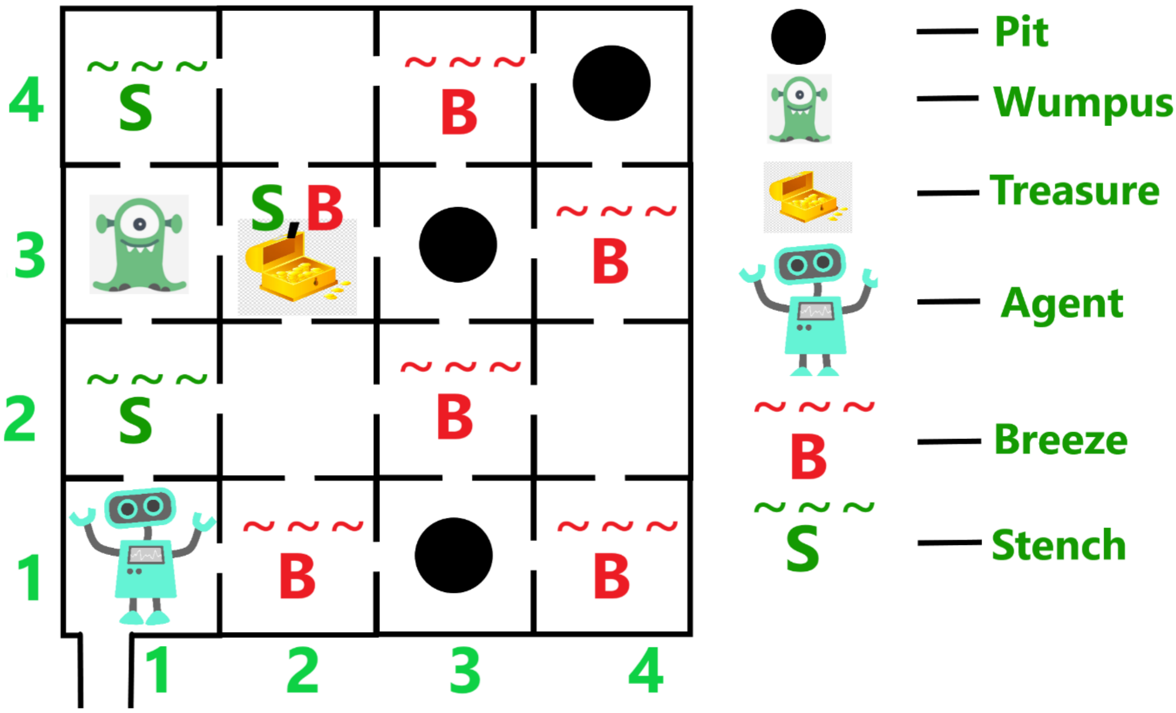
Knowledge-Based Agents: Introduction to Knowledge-Based Agents, The Wumpus World.

Knowledge-based agents (KBAs) are a type of artificial intelligence agent that uses a knowledge base to represent and reason about the world, enabling them to make informed decisions. The Wumpus World is a classic example used to illustrate how KBAs function, demonstrating knowledge representation, planning, and reasoning in a partially observable environment.

The Wumpus World is a classic example of a knowledge-based agent in AI. It involves reasoning, knowledge representation and planning. In this agent uses its knowledge of the world to make decisions and navigate safely in the given environment. In this article, we will learn more about it.

Understanding Wumpus World Problem

Wumpus World is a 4x4 grid consisting of **16 rooms**. Agent starts at Room[1,1] facing right and its goal is to retrieve treasure while avoiding hazards such as pits and the Wumpus. Agent must navigate through grid using its limited sensory input to make decisions that will keep it safe and allow it to successfully collect the treasure and exit the cave.



The Cave

Key Elements:

- **Pits**: If the agent steps into a pit it falls and dies. A **breeze** in adjacent rooms suggests nearby pits.
- **Wumpus**: A creature that kills agent if it enters its room. Rooms next to the Wumpus have a **stench**. Agent can use an **arrow** to kill the Wumpus.
- **Treasure**: Agent's main objective is to **collect the treasure** (gold) which is located in one room.
- **Breeze**: Indicates a pit is nearby.
- **Stench**: Indicates the Wumpus is nearby.

Agent must navigate carefully avoiding dangers to collect treasure and exit safely.

PEAS Description

PEAS stands for **Performance Measures**, **Environment**, **Actuators** and **Sensors** which describe agent's capabilities and environment.

1. Performance measures: Rewards or Punishments

- Agent gets gold and return back safe = *+1000 points*
- Agent dies (pit or Wumpus)= *-1000 points*
- Each move of the agent = *-1 point*
- Agent uses the arrow = *-10 points*

2. Environment: A setting where everything will take place.

- A cave with *16(4x4)* rooms.
- Rooms adjacent (not diagonally) to the Wumpus are stinking.
- Rooms adjacent (not diagonally) to the pit are breezy.
- Room with gold glitters.
- Agent's initial position - Room[1, 1] and facing right side.
- Location of Wumpus, gold and 3 pits can be anywhere except in *Room[1, 1]*.

3. Actuators: Devices that allow agent to perform following actions in the environment.

- Move forward: Move to next room.
- Turn right/left: Rotate agent 90 degrees.

- Shoot: Kill Wumpus with arrow.
- Grab: Take treasure.
- Release: Drop treasure

4. Sensors: Devices help the agent in sensing following from the environment.

- Breeze: Detected near a pit.
- Stench: Detected near the Wumpus.
- Glitter: Detected when treasure is in the room.
- Scream: Triggered when Wumpus is killed.
- Bump: Occurs when hitting a wall.

How the Agent Operates with PEAS

1. **Perception**: Agent uses sensory inputs (breeze, stench, glitter) to detect its surroundings and understand the environment.
2. **Inference**: Agent applies logical reasoning to find location of hazards. For example if it detects a **breeze**, it warns that a pit is nearby or if there's a **stench** it suspects the Wumpus is in an adjacent room.
3. **Planning**: Based on its deductions agent plans its next move avoiding risky areas like rooms with suspected pits or the Wumpus.
4. **Action**: Agent performs planned action such as moving to a new room, shooting arrow at the Wumpus or taking the treasure.

This process repeats till the agent finds the cave using its sensory inputs, reasoning and planning to achieve its goal safely.

By using **PEAS** framework agent's interactions with its environment are clearly defined, providing a structured approach to modeling intelligent behavior.

Problem Solving in Artificial Intelligence

Problem solving is a fundamental concept in artificial intelligence (AI) where systems are designed to identify challenges, make decisions and find efficient solutions. AI uses agents which are systems that perceive their environment and take actions to achieve specific goals. They go beyond simple reflex agents which react to stimuli based on pre-defined rules. Instead, problem-solving agents actively analyze situations, evaluate different options and choose the best action to reach their goal.

These agents work by:

- **Perceiving the environment:** They collect data about their surroundings such as sensor inputs or observations.
- **Defining the problem:** They clearly understand the problem including the starting point, the available actions and the desired goal.
- **Exploring different possibilities:** They consider various ways to solve the problem and evaluate which approach is likely to succeed.
- **Evaluating and deciding:** Once they explore options, they assess the outcomes and pick the best course of action based on factors like time, resources and success likelihood.
- **Learning and adapting:** Many problem-solving agents can learn from past experiences, improving their decision-making abilities over time.

Types of Problems in AI

AI systems can face different types of problems. These problems are categorized based on their impact on the system's performance and how they can be handled:

1. **Ignorable Problems**

These are minor issues that have little or no effect on the AI system’s overall performance. They are often harmless and do not require immediate attention.
Examples:

- Small inaccuracies in predictions that don't affect the final result like a slight error in image classification.
- Minor issues in data preprocessing that don't change the outcome.

Handling: Ignored as they don't interfere with the system's primary function.
2. Recoverable Problems

Recoverable problems are those where the AI system encounters an issue but can be fixed with intervention either automatically or manually, such as error-handling functions.
Examples:

- Missing data that can be filled in using statistical methods.
- System crashes that can be fixed by restoring from a backup.

Handling: Requires action like automated error handling, retraining the model or system recovery through checkpoints.
3. Irrecoverable Problems

These are severe issues that cause permanent damage or failure, making it impossible for the system to recover. They can lead to significant performance loss.
Examples:

- Corrupted training data that causes bias and reduces the model's effectiveness.
- Adversarial attacks that make the model untrustworthy.
- Overfitting, where the model becomes too specialized and cannot adapt to new data.

Handling: Requires a complete rebuild of the system including retraining the model or updating the data to eliminate underlying issues.
Steps in Problem Solving in AI

AI problem solving follows a structured, logical process similar to human thinking, it's steps include:

1. **Problem Definition:** First, the problem needs to be clearly defined. This includes understanding the inputs, constraints and what the solution should look like.
2. **Problem Analysis:** Once the problem is defined, it’s analyzed in more detail. This helps in understanding its limitations and possible solutions.
3. **Knowledge Representation:** All the important information is organized so the AI can understand and work with it. This could include creating graphs or using databases.
4. **Problem Solving:** AI uses appropriate methods to solve the problem. This often means comparing different strategies to find the most efficient one.
5. **Testing and Evaluation:** Finally, after the solution is implemented, testing and evaluation ensure the solution meets all requirements and performs as expected.

Key Components of Problem Formulation in AI

Effective problem-solving in AI is dependent on several important components:

1. **Initial State:** This is the starting point of the problem where the AI begins its process. It sets the context and helps identify how the agent will approach the challenge.
2. **Action:** At this stage, AI identifies all the possible actions it can take from the initial state. Each action has an impact on how the system moves closer to solving the problem.

3. **Transition:** This refers to how the system changes from one state to another after an action is taken. Transition modeling helps show how the agent's actions influence the next steps.
4. **Goal Test:** Once an action is taken, AI checks if it has reached its goal. If the goal is achieved, the problem-solving process stops and the solution is considered complete.
5. **Cost Function:** This step assigns a numerical value to the cost of achieving the goal. The cost can include resources like time, energy or money and helps decide the most efficient way to reach the goal.

Techniques for Problem Solving in AI

Several techniques are available to solve different types of problems in AI. The choice of technique depends on the nature of the problem and the resources available.

1. **Search Algorithms**

Search Algorithms are important for exploring possible solutions. There are two main categories:

- **Uninformed Search**: These algorithms such as breadth-first and depth-first search, don't use additional knowledge about the problem and explore all possible paths.
- **Informed Search**: Algorithms like A* use heuristics or domain-specific knowledge to guide the search toward more promising paths making the search process more efficient.

2. Constraint Satisfaction Problems (CSP)

In CSPs the goal is to find solutions that satisfy a set of constraints. Techniques like backtracking, constraint propagation and local search are used to explore the solution space efficiently while ensuring that all constraints are met.

3. Optimization Techniques

AI often addresses optimization problems where the goal is to find the best possible solution among many options. Techniques include:

1. Linear programming for solving problems involving linear relationships.
2. Dynamic programming for breaking down complex problems into smaller subproblems.
3. Evolutionary algorithms like genetic algorithms that simulate natural selection to find optimal solutions.

4. **Machine Learning**

Machine learning allows AI systems to learn from data and improve over time. The three major learning types are:

1. Supervised Learning: Learning from labeled data to make predictions or classifications.
2. Unsupervised Learning: Discovering patterns in data without labeled outputs.
3. Reinforcement Learning: Learning through trial and error, receiving feedback in the form of rewards or penalties.

5. **Natural Language Processing (NLP)**

Natural Language Processing (NLP) helps AI systems understand and generate human language. Techniques like tokenization, Part-of-speech (PoS) and named entity recognition are important for chatbots, language translation and text summarization.

Advantages of AI in Problem Solving

1. **Efficiency**: It can solve problems much faster than humans, processes large amounts of data and identifying patterns quickly.
2. **Accuracy**: With the right data and models, AI can make decisions with high accuracy, especially in tasks like medical diagnoses or image recognition.
3. **Automation**: It automates repetitive tasks, freeing up human time for more complex and creative work.
4. **Adaptability**: AI systems, particularly those using machine learning can improve over time, adapting to new data and evolving challenges.
5. **Cost-Effectiveness**: Once implemented, it can reduce costs by simplifying operations, improving productivity and minimizing human error.

Challenges in AI Problem Solving

1. **Complexity**: Some problems are highly complex, requiring significant computational resources and time to solve.
2. **Data Quality**: AI systems depend on data quality and poor or biased data can lead to inaccurate or misleading outcomes.
3. **Interpretability**: Many AI models, especially deep learning, lack transparency, making it hard to understand how decisions are made.
4. **Ethics and Bias**: AI can conserve biases in data which leads to unfair or unethical outcomes, ensuring fairness is a key challenge.

As AI continues to evolve, enhancing problem-solving capabilities will remain important for advancing technology and improving human experiences.